



The role of IT leadership in realizing the return on investment of AI coding assistants.

by

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Abstract

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Background

Artificial intelligence (AI) is one of the important structural influencers on productivity growth and competitive advantage today (Gillespie et al., 2025). As of 2023, generative AI systems have progressed from experimentation to operationalization within organizational processes (Gillespie et al., 2025). Recent global studies suggest that organizations are investing more in AI with clear expectations of understanding tangible economic gains, cost savings, and performance improvements (Gillespie et al., 2025). Research on information systems and management has advanced from adoption to value understanding, with a focus on distinguishing organizational value from investment in AI technologies as a function of organizational governance and management capability rather than technological capability (Hossain et al., 2025).

In software engineering, generative AI tools represent one of the most prominent manifestations of AI-enabled augmentation (Peng et al., 2023). Providing developers with functionalities such as code suggestions and entire code functions in real time based on context, documentation, and debugging (Peng et al., 2023). Practical controlled experiments provide conclusive evidence that AI-assisted developers complete software development tasks faster and report increased productivity of about 55.8% compared with control groups (Peng et al., 2023). Collaboration between humans and AI systems is a paradigm shift from the traditional concept of automation, in which AI enhances developers' capabilities rather than replacing them with machines (Peng et al., 2023). Initial performance improvements demonstrated in controlled promising experiments may not necessarily guarantee long-term organizational performance benefits. Empirical studies report considerable variation in the productivity gains from AI-assisted software development tasks across software development teams and organizations.

Both organizational culture and leadership have traditionally been acknowledged as factors that impact the success of technology implementation. In the digital domain, it is the values and

norms that foster a culture of experimentation, learning, and data-driven decision-making, as well as a willingness to adopt technological changes. New studies have found that digital culture reinforces the association between human AI competencies and innovation outcomes, suggesting that cultural readiness moderates the performance effects of AI investments (An et al., 2024). Likewise, the same is true for employee engagement in AI initiatives, ethical governance, and leadership's role in influencing strategic alignment. While the importance of digital culture and leadership is increasingly acknowledged in the AI adoption domain, the current body of research remains dispersed (An et al., 2024). Most studies analyse AI adoption from a macro-organizational perspective or focus on individual productivity effects, failing to consider the integration of digital culture, leadership, and ROI in software engineering (Peng et al., 2023). ROI is increasingly becoming the sole rationale for investing in enterprise AI initiatives.

African organizations are accelerating the adoption of AI technologies to address structural inefficiencies and resource limitations. Emerging market organizations have reported high levels of workplace AI experimentation and positive attitudes toward AI productivity (Gillespie et al., 2025). However, the extant literature also emphasizes the lack of digital skills pipelines, governance maturity, and leadership ability, among other things, as difficulties associated with the adoption of GenAI technologies (Hughes et al., 2026). In this regard, the importance of organizational culture and leadership cannot be overstated. In the absence of strategic direction, trust, and a learning-oriented culture, the adoption of AI technologies is more inclined toward fragmentation and symbolic adoption than toward creating value. The digital transformation landscape in Africa is a valid environment for exploring the impact of socio-organizational factors on the achievement of returns from AI technologies.

In the context of software engineering in South Africa, it provides a relevant setting for understanding how organizations embrace and use advanced AI tools. Research indicates that AI is emerging as a key factor influencing organizational change and shaping work practices and culture as organizations increasingly embrace AI tools across their business processes and operations (Murire, 2024). There are opportunities and challenges to address to leverage organizational performance through AI adoption and alignment with organizational culture and structure, as indicated by research reviews on AI tool adoption in organizations (Romeo & Lacko, 2025). Though the adoption of AI technology affects many aspects of the organization's performance positively, like productivity and output, the effect of AI technology on aspects like

the organization's culture and employees' morale and engagement is complicated and depends on many factors, as indicated by the study on the impact of AI tools in workplace settings (Kassa & Worku, 2025). Studies on the effect of generative AI tools on software engineering have indicated that AI technology positively affects organizations by enhancing productivity in coding and other software engineering activities through employees' adoption of AI tools (Alenezi & Akour, 2025).

Leadership and organizational culture play an important role in the interpretation, adoption, and usage of AI technologies by work teams. Research on AI adoption has found that AI-related technology adoption is also dependent on leadership styles that facilitate embedding technology initiatives into organizational culture (Romeo & Lacko, 2025). Research on the adoption of AI-related technology has also found that the adoption of AI-related technology is dependent on the adoption of the organizational culture that is conducive to the adoption of the technology (Kassa & Worku, 2025). Organizational culture research further indicates that cultural factors, specifically norms related to innovation, collaboration, and adaptability, modulate how employees engage with AI tools and integrate them into their work (Murire, 2024). In software engineering teams, where AI coding assistants alter routines and task assignments, leadership and culture therefore likely determine the extent to which AI investments yield returns in productivity, quality, and financial outcomes, making them essential variables in understanding realized returns on investment.

Problem Statement

Generative AI has shifted its focus from experiment deployment to integrating it into the core knowledge work process. Substantial experimental results show that generative AI can significantly improve the productivity and quality of cognitive work outcomes, especially among lower-performing members, thereby affecting the performance distribution within the organization (Noy & Zhang, 2023). Field experiment results show that the application of generative AI in the real world can improve the organization's productivity and influence relevant outcomes, such as task completion and retention, indicating that the ROI on generative AI comes from its application within the organization, not the technology itself (Brynjolfsson et al., 2025). In the software development field, three large-scale field experiments have shown that providing developers with GitHub Copilot can improve task completion and development activities, providing empirical support for the ROI of generative AI in coding assistance (Cui et al., 2026). Controlled experiment results show that developers using AI coding assistance complete programming tasks more quickly and at a higher rate, providing a business case for AI coding assistance tools based on productivity (Peng et al., 2023).

Despite these measurable improvements, existing research evidence consistently shows that the performance impact differs across contexts. Research in software engineering has shown that AI-assisted code generation produced by GitHub Copilot improves speed but also creates variability in correctness, maintainability, and review effort; this shows the value to depend on the management and integration of such tools and systems (Moradi Dakhel et al., 2023). Research on AI governance in organizations has argued that to sustain value, organizations must have structured processes and systems to ensure clear accountability and supervisory leadership to transform experimentation into sustained performance improvement (Mäntymäki et al., 2022). Research on organizational innovation with generative AI has also shown that organizations can obtain different performance impacts depending on whether AI is used for

automation or augmentation; this shows the role of strategic leadership and organizational culture in determining the benefits obtained (Holmström & Carroll, 2025). Overall, this existing research evidence shows the role of organizational culture and strategic leadership in determining whether the improvements in productivity due to AI coding assistants can be sustained and converted to measurable return on investment.

The relevance of these mechanisms can be further accentuated in the context of Africa since organizations face structural limitations in skills development, governance, and digital sophistication. An empirical study of AI readiness in Africa identified the level of preparedness and the institutional ability to adopt AI, thus emphasizing the role of organizations in the transition from AI pilot projects to performance results (Isagah & Ben Dhaou, 2024). In the context of South Africa, empirical studies of the financial sector have identified the role of organizational factors such as leadership support and organizational readiness in the adoption of new and emerging technologies, thus emphasizing the relevance of these factors to the achievement of ROI in AI coding assistants in software development teams (Matsepe & Van der Lingen, 2022). Empirical research carried out within the context of South Africa suggests the use of sociotechnical systems in AI adoption, as well as the importance of change management for the accomplishment of performance outcomes related to AI adoption (Smit et al., 2024). Another empirical research within the context of South Africa reveals the influence of managerial perceptions regarding AI adoption on the accomplishment of performance outcomes within the context of human resource management as a field of AI application.

Collectively, the global evidence base shows that AI coding assistants can improve productivity, and the organizational evidence base shows that sustained ROI depends on alignment, effective governance, and culture. The evidence from Africa and South Africa also shows that context plays an important role in the success of technology adoption. Therefore, the evidence base provides justification for an in-depth investigation to explore the role of organizational culture and leadership in the achievement of ROI with AI coding assistants in software development teams in South Africa.

Research Objectives

0.1 Main Research Objective

To develop an integrated framework linking financial metrics, organizational culture, and leadership to assess and enhance the return on investment of AI coding assistants in South African software engineering teams.

0.1.1 Sub Research Objectives

1. To identify and operationalize the financial costs and benefits that define return on investment for AI coding assistants.
2. To examine how organizational culture attributes influence productivity and quality outcomes from AI coding assistants.
3. To assess which leadership practices most strongly affect return on investment outcomes.
4. To apply and validate an integrated return on investment framework within large South African organizations.
5. To formulate practical recommendations that enable managers to maximize return on investment through cultural and leadership interventions.

Research Question(s)

0.2 Main Research Question

How do organizational culture and leadership shape the return on investment of AI coding assistants in South African software engineering teams?

0.2.1 Sub Research Questions

1. What financial costs and benefits define return on investment for AI coding assistants in software engineering teams?
2. How do organizational culture attributes influence productivity and quality outcomes from AI coding assistants?
3. Which of the leadership behaviours strongly affect return on investment outcomes from AI coding assistants?
4. How can financial and organizational factors be integrated into a practical return on investment framework for large South African organizations?
5. What practical guidance can be offered to managers to maximize returns from AI coding assistance investments through cultural and leadership interventions?

Keywords: AI, ROI, GenAi.

Declaration

I, Celumusa H Zulu, certify that the dissertation submitted by me for the Master of Commerce in Information Technology Management degree at the University of Johannesburg is my independent work and has not been submitted by me for a degree at another university.

CELUMUSA H ZULU

(No signature required)

Dedication

To my Mother,
who always believe in me,
and to all my Siblings
who always believe in me

Acknowledgements

First and foremost, infinite gratitude and appreciation to my Creator, Who has given me the will to start studying again and the endurance to see this through in a time when my world was turned on its head. My perseverance, accomplishments and very being are only by Your will.

You are the very foundation my success is built on.

My deepest gratitude to my Mother, Sister and my brother's as well for always remembering me in your prayers.

My sincerest appreciation to my Supervisor Dr. Siyabonga Mhlongo, who has always been my light through this journey. You got me started on this, thank you for seeing me through to the finish line.

I'd also like to extend my sincerest appreciation to all the people that took their precious time and completed the survey to make this study possible.

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Chapter 1

Introduction

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1.1.2 Some maths

L^AT_EX is very good at presenting mathematics.

Inline equation : $ax^2 + bx + c = 0$

Display equation:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Numbered equation:

$$\frac{\partial}{\partial t} U + \nabla \cdot F = 0 \tag{1.1}$$

Aligned equation (not how equals signs line up):

$$\begin{aligned} \mathbf{a} \cdot \mathbf{b} &= \sum_{i=1}^n a_i b_i \\ &= a_1 b_1 + a_2 b_2 + \cdots + a_n b_n; \end{aligned} \tag{1.2}$$

Aligned equation with no numbers:

$$\begin{aligned} \mathbf{a} \cdot \mathbf{b} &= \sum_{i=1}^n a_i b_i \\ &= a_1 b_1 + a_2 b_2 + \cdots + a_n b_n; \end{aligned}$$

1.1.3 Theorems, proofs, definitions and examples

Theorem 1.1 (Fermat's last theorem). *No three positive integers a , b , and c satisfy the equation $a^n + b^n = c^n$ for any integer value of n greater than 2.*

Proof. Left as an exercise for the reader. □

Definition 1.1. *The intersection of two sets A and B , denoted by $A \cap B$, is the set of all objects that are members of both the sets A and B .*

Example 1.1

Given the two sets $A = \{x : x \in \mathbb{N}, x < 5\}$ and $B = \{x : x \in \mathbb{N}, x \text{ is even}\}$ then $A \cap B = \{2, 4\}$.

1.2 Figures and tables

Figure 1.1: This is a figure caption, note how it appears underneath the figure.

Table 1.1: This is a table caption, not how it appears above the table.

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1.2.1 Program code

Listing 1.1: A MATLAB function to compute the first n numbers of the Fibonacci series

```
function y = fibonacci(n)

% This function calculates the first n terms in the Fibonacci series

y(1) = 0;
y(2) = 1;

for i = 3 : n
    y(i) = y(i-2) + y(i-1)
end

end
```

1.3 Referencing

References can be cited so that the author names(s) are a part of the sentence, e.g., Mhlongo et al. (2023), Ofusori et al. (2024), or Achuthan and Khobragade (2026).

Alternatively, references can be cited so that the author names appear in the brackets (for when the name of the author is not relevant to the sentence), e.g., (Mhlongo et al., 2023), (Ofusori et al., 2024). You can also use the `\autocite` command for end of sentence citations (Achuthan & Khobragade, 2026; Mhlongo et al., 2023; Ofusori et al., 2024).

For additional information on citation commands, consult the `biblatex` cheat sheet via this URL: <https://tug.ctan.org/info/biblatex-cheatsheet/biblatex-cheatsheet.pdf>. The comprehensive `biblatex-apa` archive can be found here: <https://ctan.org/pkg/biblatex-apa>.

Cross referencing is easily done by using the label of the item you are referencing to (see source code for details).

- Chapter 1
- Section 1.1
- Equation (1.1)
- Table 1.1
- Figure 1.1
- Page 2

Above is an example of a bulleted list. You can also create numbered lists:

1. First list item
2. Second list item
 - (a) First sub list item
 - (b) Second sub list item
3. Third list item

1.4 List of Acronyms

Acronyms can be introduced using the `\gls{}` command. This works in conjunction with what is included in the `acronyms.tex` file. For example, **my simple acronym (MSA)** is an entry in this file, and **MSA** can be used throughout the document. See <https://www.overleaf.com/learn/latex/Glossaries> for more usage details.

Chapter 2

Literature Review

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Chapter 3

Research Methodology

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Chapter 4

Results

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Chapter 5

Discussion

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Chapter 6

Conclusion

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Appendix A

Ethical Clearance Certificate



CBE RESEARCH ETHICS COMMITTEE

Dear Researchers

ETHICAL APPROVAL GRANTED FOR RESEARCH PROJECT

Decision: Clearance granted

This letter serves to confirm that the proposed research project indicated in the table below, has been reviewed by the Department of Applied Information Systems at the University of Johannesburg. Ethical clearance is hereby granted and is valid for three years, from 01 January 1900 until 31 December 2999.

Applicant	
Supervisor	
Student/staff number	
Title	
Decision date at meeting	
Reviewers	
Ethical clearance code	
Rating of application	

CODE 01 – Approved

CODE 02 – Approved with suggestions/requirements
with no re-submission

CODE 03 – Referred back

CODE 04 – Disapproved, cannot re-submit

The researcher/s may now commence with the study providing that:

1. The researcher/s will ensure that the project adheres to ethical research requirements
2. The researcher/s will be conducting the study as set out in the approval application
3. The researcher/s will ensure that project adheres to all applicable legislation, scopes of practice, professional codes of conduct and scientific standards as it pertains to the field or study.
4. The researcher/s will bring under the attention of the research ethics committee any proposed changes, concerns that arise, and unexpected ethical management issues
5. Any changes that can affect the study-related risks for participants or researchers must be reported to the committee in writing
6. Only data collected during the three valid period is permissible and valid for the study

7. No data collect before or after the three valid period is ethically valid and the Research Ethics Committee takes no responsibility for any issues arising in this situation
8. No fieldwork activities may continue after the ethical clearance has expired. A request for an extension of ethical clearance can be made in writing to the REC
9. All data collection must be done within the Government regulations of the lockdown
10. The researcher/s must be aware that COVID-19 may influence the responses and needs to be controlled for

Furthermore, the researcher/s must please address the following comments from the ethics committee:

- N/A

Prof M Name

AIS REC

aisrec@uj.ac.za

Appendix B

Appendix Chapter

B.1 Appendix section

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Appendix C

Another Appendix Chapter

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